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Reusability report: Uncovering associations in biomedical bipartite networks via a bilinear attention network with domain adaptation

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Conditional domain adversarial learning presents a promising approach for enhancing the generalizability of deep learning-based methods. Inspired by the efficacy of conditional domain adversarial networks, Bai and colleagues introduced DrugBAN, a methodology designed to explicitly learn pairwise local interactions between drugs and targets. DrugBAN leverages drug molecular graphs and target protein sequences, employing conditional domain adversarial networks to improve the ability to adapt to out-of-distribution data and thereby ensuring superior prediction accuracy for new drug–target pairs. Here we examine the reusability of DrugBAN and extend the evaluation of its generalizability across a wider range of biomedical contexts beyond the original datasets. Various clustering-based strategies are implemented to resplit the source and target domains to assess the robustness of DrugBAN. We also apply this cross-domain adaptation technique to the prediction of cell line–drug responses and mutation–drug associations. The analysis serves as a stepping-off point to better understand and establish a general template applicable to link prediction tasks in biomedical bipartite networks.

In the realm of biomolecular systems, a plethora of interaction/association bipartite networks can be found[1]. Identifying previously undiscovered associations within biomedical networks plays a pivotal role in unravelling the intricate molecular mechanisms underlying complex human diseases[2]. Owing to the expansive space of biomedical entities, real-world applications frequently entail predicting association pairs that are unseen and dissimilar to any pairs present in the training data. A robust model should possess the capability to transfer acquired knowledge to a new domain characterized by different distributions[3]. Recently, Bai et al. introduced a model for drug–target prediction in the field of biomedicine[4]. In their method, known as DrugBAN, the encoded local representations are fed into a bilinear attention network (BAN) integrated with domain adaptation. A conditional domain adversarial network (CDAN)[5] was leveraged to enhance cross-domain generalization capabilities beyond the learned distribution.

In this study our aim is to assess the reproducibility of the results presented by Bai and colleagues[4]. We also highlight the feasibility of using different clustering-based pair splitting strategies and showcase the applicability of the cross-domain adaptation approach in other biomedical contexts: the prediction of cell line–drug responses and mutation–drug associations.

Reproducing the reported results

We obtained the source code of DrugBAN from the repository at https://github.com/peizhenbai/DrugBAN and proceeded to retrain the models using the provided datasets following the methodology

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Source	Strategy	Setting	Performance (AUROC) BindingDB random	BioSNAP random	BindingDB cluster	BioSNAP cluster
Raw	CDAN		NA	NA	0.6176	0.6913
Raw	Non CDAN		0.9612	0.9064	0.5949	0.6354
Raw	CDAN	Max pooling	NA	NA	0.7360	0.7013
Raw	Non CDAN	Max pooling	0.9614	0.9278	0.5741	0.6317
Resplit	CDAN	Seed 1	NA	NA	0.5693	0.6663
Resplit	Non CDAN	Seed 1	0.9576	0.9035	0.5582	0.6217
Resplit	CDAN	Seed 2	NA	NA	0.6269	0.7173
Resplit	Non CDAN	Seed 2	0.9584	0.9056	0.5455	0.6293
Resplit	CDAN	Seed 3	NA	NA	0.6228	0.7004
Resplit	Non CDAN	Seed 3	0.9749	0.9477	0.5217	0.6152
Resplit	CDAN	Seed 4	NA	NA	0.5660	0.7004
Resplit	Non CDAN	Seed 4	0.9634	0.9047	0.5236	0.6152
Resplit	CDAN	Seed 5	NA	NA	0.7063	0.6443
Resplit	Non CDAN	Seed 5	0.9630	0.8986	0.5902	0.5874

Fig. 1 | Reproducing the main reported performance results of DrugBAN. We conducted five rounds of random dataset splitting using different random seeds and report their performance (AUROC values, colour scale) on the newly split dataset. Resplit refers to the process of resplitting the dataset using random seeds. CDAN denotes the embedding of the CDAN module in the cross-domain evaluation scenario and non CDAN indicates the absence of the CDAN module either in the cross-domain or in the intradomain evaluation scenario. Max pooling indicates the modification made to the model's pooling method. Seed1–Seed5 represent the five different random seeds used for dataset splitting. NA, not applicable to this study.

described by Bai et al.[4]. Within this methodological framework, DrugBAN integrates a CDAN to improve the ability to generalize from the source domain with labelled data to a target domain where only unlabelled data exist. Leveraging the CDAN mechanism, the authors designed a simultaneous alignment of the joint representation and the predicted classification distributions across the source and target domains. This alignment is facilitated by conditioning the domain discriminator. The integration of adversarial learning markedly improves the model's capability to identify potential associations in the target domain, using the learned representations from the source domain. This is particularly effective for new drugs or proteins, as it allows the framework to make informed predictions even in the absence of direct interaction data.

To ensure the reliability of our results, we performed five independent runs with different random seeds for each experiment and reported the average performance. The average area under the receiver operating characteristic curve (AUROC) values we obtained for all three datasets closely aligned with those reported in the original paper[4], with deviations of 0.01 from their reported values (Fig. 1; the performance on the Human dataset[6] is shown in Supplementary Fig. 1). These deviations may be attributed to inconsistent versions of certain libraries within the Anaconda environment on Linux. In the constructed cross-domain scenario, we also conducted experiments in BindingDB[7] and BioSNAP[8] to compare the classification performance of the model with and without the CDAN module. We observed that the best epoch reported occurred within the first few iterations out of 100, suggesting that the model struggles to learn in the cross-domain scenario with CDAN, aligning with the original paper's conclusion. Next, we performed five independent random splits of the original dataset using different random seeds, following the two segmentation strategies mentioned in the original paper. The results obtained from the re-segmented datasets were similar to those reported in the original paper. Compared with in-domain predictions, all models experienced a significant drop in performance due to the reduced information overlap between training and test data, but the inclusion of a domain adaptation module led to significant improvements.

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Source	Strategy	Performance (AUROC) BindingDB	Performance (AUROC) BioSNAP
Raw	CDAN	0.6716	0.6913
Raw	Non CDAN	0.5949	0.6354
R-cluster	CDAN	0.6312	0.6453
R-cluster	Non CDAN	0.5249	0.6150
K-cluster	CDAN	0.5661	0.7000
K-cluster	Non CDAN	0.5295	0.6563

Fig. 2 | Cross-domain performance of the DrugBAN method based on re-clustering. The dataset was re-clustered and split using both hierarchical clustering and *K*-means algorithms to assess the performance of DrugBAN in the newly constructed cross-domain scenarios. R-cluster and *K*-cluster represent the results based on the original hierarchical clustering and *K*-means algorithms, respectively.

Re-clustering the raw dataset

The proposed DrugBAN framework demonstrates strong domain adaptability, leveraging the CDAN module for cross-domain prediction. By transferring learned knowledge from the source domain to the target domain, the model enhances its cross-domain generalizability. It should be noted that the construction of the cross-domain scenario relies on clustering drug compounds and target proteins, making the clustering results crucial for cross-domain evaluation. To test the robustness of the CDAN module and cluster method, we re-clustered the drug compounds and target proteins separately in BindingDB and BioSNAP using the original method and *K*-means algorithms[9]. The statistics of drug clusters and target protein clusters are shown in Supplementary Table 1.

The numbers of drug clusters and protein clusters obtained in BindingDB and BioSNAP exhibited slight variations from those reported in the original paper[4], potentially due to uncertainties in the feature extraction process. Different clustering algorithms and data splitting methods contributed to the generation of distinct source and target domains. The experiments were then conducted using the re-clustered datasets as described in Fig. 2, with deviations of 0.04 from their reported values. Specifically, the experimental results based on *K*-means clustering showed lower performance than the original results on BindingDB. Interestingly, on BioSNAP, the *K*-means-based results demonstrated higher performance (row 5 versus row 1 in Fig. 2). These results demonstrate the strength of the CDAN module in generalizing prediction performance across domains and its robustness to clustering methods.

Application of DrugBAN beyond the raw dataset

To further investigate the reusability of DrugBAN, we undertook additional experiments to apply DrugBAN to the Davis and KIBA datasets[10,11], part of the updated benchmark collection assembled by Wu et al.[12] for drug–target affinity (DTA) prediction. We standardized the datasets, stratifying drug–target pairs into affinity and non-affinity categories on the basis of median affinity values. The statistics of the refined DTA benchmark datasets are summarized in the Supplementary Table 2. Using the DrugBAN model trained on original distinct datasets (BindingDB and BioSNAP), we observed significant differences in drug–target affinities across categories (Fig. 3), confirmed by the Wilcoxon rank sum test (P value <0.05). The y axis represents drug–target affinity. A value of 1 represents the group predicted to have drug–target interaction and 0 represents the group predicted to have no drug–target interaction. This divergence underscores the DrugBAN model's predictive validity. These experiments demonstrate DrugBAN's high accuracy and potential generalization capabilities on novel datasets, offering a foundational tool for advancing drug discovery and development.

Application of DrugBAN to various bipartite networks

In their original paper, Bai et al.[4] showed the generality of their proposed ideas, which aimed to mitigate the domain distribution shift between the source and target domains. These ideas may be effectively extended to address other interaction prediction problems. To further assess the utility and robustness of this pipeline, we conducted experiments on other two significant biological problems: the prediction of cell line–drug responses and mutation–drug associations.

For cell line–drug response prediction, we carefully extracted 517 cell lines with response data against 279 drugs. The drug response data were quantified using half-maximum inhibitory concentration (IC_{50}) values[13], measured in micromolar concentrations. Lower IC_{50} values signify greater sensitivity of the cell line towards the drug, whereas higher IC_{50} values indicate the opposite. The statistics of our drug response dataset are summarized in Supplementary Table 3. We applied DrugBAN to drug molecular graphs and copy number variation profiles of the cell lines for in-domain and cross-domain prediction. In the context of in-domain prediction, the model demonstrated promising performance with an AUROC exceeding 0.98, which does not necessarily reflect the real-world performance of the model in prospective applications. As highlighted in ref. 6, the observed high accuracy could be attributable to a hidden ligand bias, indicating that drug features may play a key role in drug response prediction, rather than interaction patterns.

We therefore investigated the cross-domain performance of DrugBAN in drug response prediction, comparing its performance with and without the CDAN module. The results of the cross-domain prediction are depicted in Fig. 4a, which indicates that the models have a performance drop from in-domain to cross-domain prediction. However, we can see that DrugBAN with the CDAN module still achieved superior performance to the version without CDAN. The prediction performance in terms of receiver operating characteristic and precision recall curves can be observed in Supplementary Fig. 2. Remarkably, both DrugBAN with and without the CDAN module exhibited promising performance, as evidenced by the AUROC of 0.8047 with CDAN, surpassing the performance without the CDAN module by 3.35%.

For mutation–drug association prediction, we meticulously curated a dataset comprising 661 mutations and their corresponding interactions with 184 drugs. The classification of the mutation–drug pairs was approached as a binary problem, distinguishing between sensitivity and resistance. The statistics of our mutation–drug association dataset are summarized in Supplementary Table 4. We used DrugBAN for this prediction, incorporating drug Simplified Molecular Input Line Entry System and mutation data. We also investigated the cross-domain performance of DrugBAN in mutation–drug association prediction, comparing its performance with and without the CDAN module. The results are depicted in Fig. 4b. We can see that DrugBAN with the CDAN module achieved superior performance to the version without CDAN, as evidenced by the AUROC of 0.7475 with CDAN, surpassing the performance without CDAN by 12.66%. These findings are

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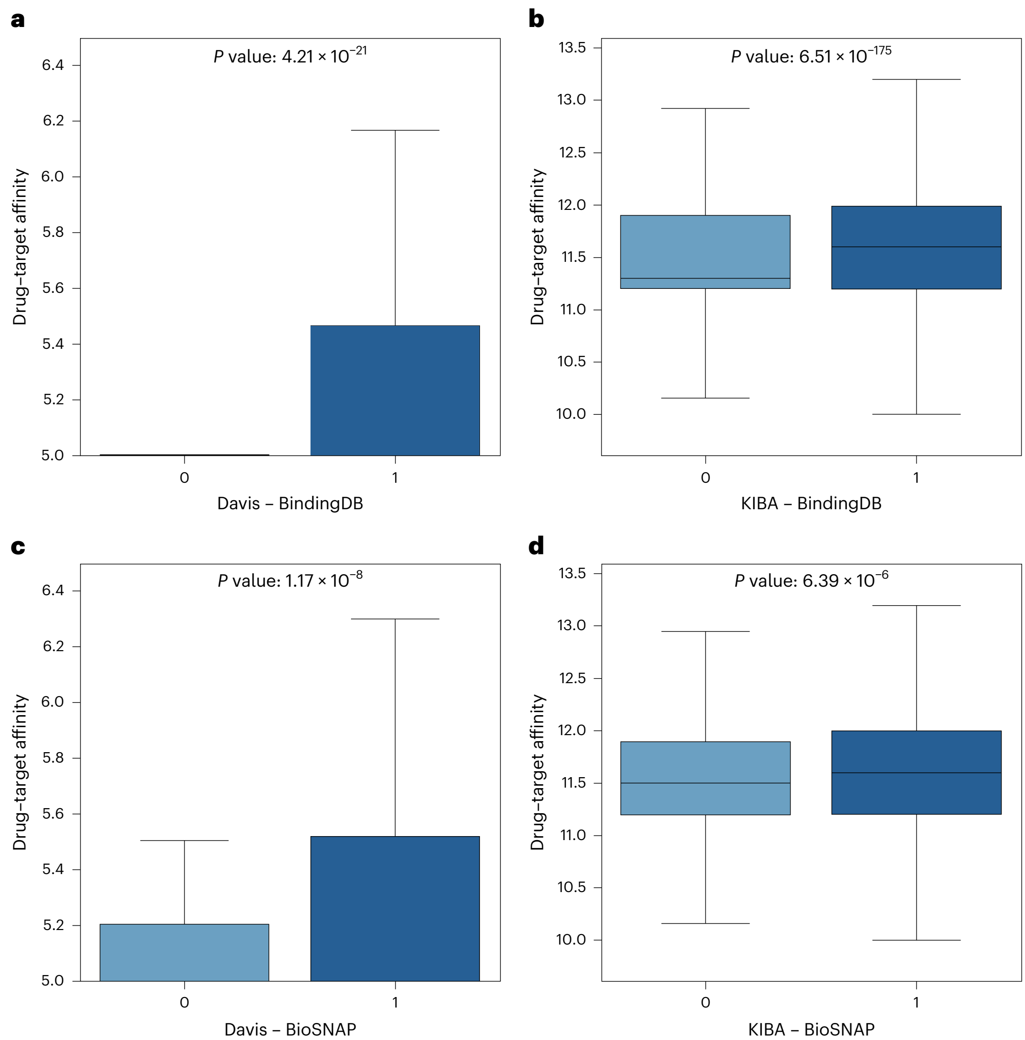

Fig. 3 | Cross-domain performance of the DrugBAN method on other drug–target datasets. a, The model trained on BindingDB for the Davis dataset. **b**, The model trained on BindingDB for the KIBA dataset. **c**, The model trained on BioSNAP for the Davis dataset. **d**, The model trained on BioSNAP for the KIBA dataset. The *P* values were calculated using the Wilcoxon rank sum test. The value of 1 denotes the group predicted to have drug–target interaction, whereas 0 signifies the group predicted to lack drug–target interaction. The box plots visually represent the median, the first quartile (Q1), third quartile (Q3) and the whiskers, which are calculated as 1.5 times the difference between Q3 and Q1.

consistent with the results reported in the original paper[4], confirming that the CDAN enhances the generalizability of DrugBAN.

Discussion and future directions

In this study we successfully replicated a significant portion of the main results reported by Bai et al.[4] by utilizing their code and data. This replication corroborates the reliability and reproducibility of their findings. We also explored the implementation of an early stopping strategy in our model. By saving the best-performing parameters and results from 100 epochs, our objective was to strike a balance between optimizing the model's learning effectiveness and conserving computational resources. Experimental results demonstrated that incorporating the early stopping strategy led the model to converge to a local optimum. It is worth noting that changing the model's pooling method did not yield substantial performance changes, suggesting that DrugBAN is not highly sensitive to hyperparameter settings.

The cross-domain generalization capability of DrugBAN holds immense value and biological significance in real-world applications. It enables accurate predictions for new drugs or proteins, even when they are absent from the training set. This capability could greatly benefit drug combination design and clinical applications. We therefore dedicated efforts to exploring this aspect. First, following the general set-up for domain adaptation, we leveraged the original clustering results to perform five different random splits of the source and target domains. This was done to verify the model's cross-domain generalizability, and the experimental results showcased its remarkable performance. We also conducted a recalculation of the features of drug compounds and proteins in the BindingDB and BioSNAP datasets, respectively. Subsequently, we employed various clustering algorithms to obtain distinct clusters for both drugs and proteins. These clusters were randomly selected in proportion to establish the source and target domains, thereby creating new cross-domain scenarios. We conducted cross-domain evaluation of DrugBAN and observed a significant improvement in its performance when trained with the CDAN module. These findings provide strong evidence that the CDAN effectively transfers learned knowledge from the source domain to the target domain, thereby enhancing the model's generalization performance.

In addition to reproducing the results of Bai et al.[4], we verified the robustness of DrugBAN in terms of changing the prediction task (from drug–target to drug–cell line or drug–mutation). Further analysis of our results revealed the strong performance of the DrugBAN model in the prediction of cell line–drug responses and mutation–drug

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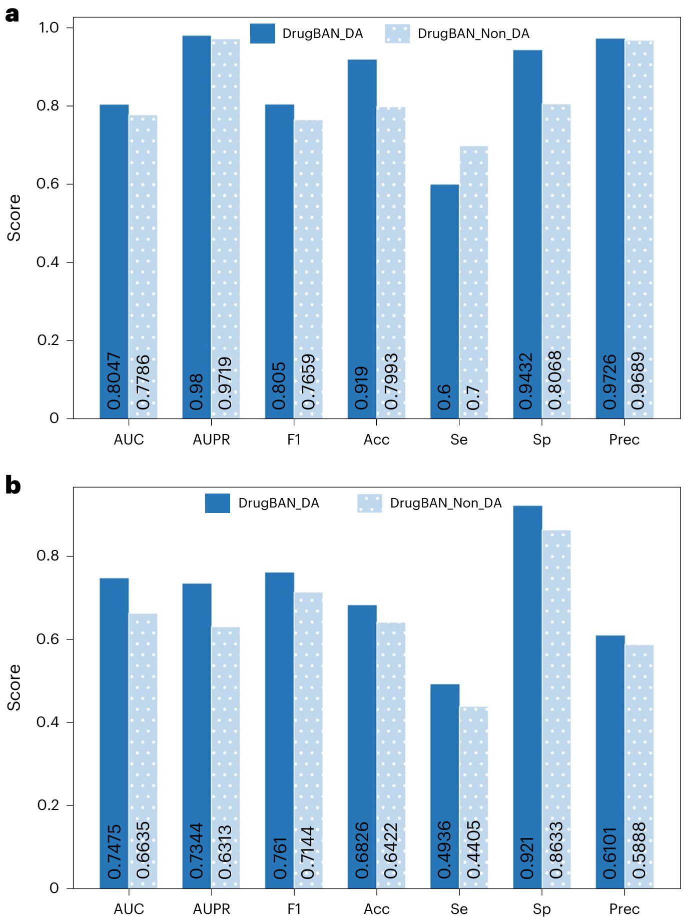

Fig. 4 | Cross-domain performance of DrugBAN method in other biomedical context. a, Cell line–drug response prediction. **b**, Mutation–drug association prediction. F1, Acc, Se, Sp and Prec represent the F1 score, accuracy, sensitivity, specificity and precision, respectively. AUC, area under the receiver operating characteristic curve; AUPR, area under the precision-recall curve. DrugBAN_DA represents the DrugBAN version with the CDAN module. DrugBAN_Non_DA represents the DrugBAN version without the CDAN module.

associations. This indicates that DrugBAN has the capability to extract relevant features from drug molecular graphs and genetic profiles, enabling accurate prediction of various biomedical problems.

In summary, reproducing the reported results has provided valuable insights into the reliability of the DrugBAN model. Employing DrugBAN in drug response prediction and drug–mutation association prediction showed that DrugBAN has good scalability and transferability, and can be applied to various biomedical issues. However, there are several avenues for future exploration and improvement. With the abundance of biological networks and a wide range of intriguing prediction targets, it would be instructive to conduct further systematic studies on the expansion of this algorithm to address prediction problems in other heterogeneous networks[14]. Future research should focus on enhancing model generalization to establish general templates that can be effectively applied to a wide range of biomedical challenges.

Methods

Reproducibility of the original results

We evaluated the performance of DrugBAN on three public datasets: BindingDB, BioSNAP and Human. For Bindingdb and BioSNAP, we conducted in-domain and cross-domain evaluations using a random split strategy and a cluster-based split strategy, respectively. According to the DrugBAN GitHub repository, the code to reproduce the reported results is easy to execute.

Re-clustering strategy

Given that the construction of the cross-domain scenario relies on clustering drug compounds and target proteins, we re-clustered the dataset from the original paper to verify the robustness of the method. For this purpose, we implemented the hierarchical clustering algorithm with single linkage[15] and *K* means. ECFP4 features[16], which are extended-connectivity fingerprints capturing structural information of drug compounds up to four bonds away from each atom, were used to represent drug compounds. Meanwhile, pseudo amino acid composition (PSC) integrates protein properties and structure into traditional amino acid composition through pseudo components, employed for target proteins. Jaccard distance and cosine distance were applied to ECFP4 and PSC, respectively, with a minimum distance threshold of $\gamma = 0.5$. This threshold selection ensured that clusters were not excessively close, enabling the creation of meaningful cross-domain scenarios.

Drug response prediction using DrugBAN

To predict cell line–drug responses, we curated information from the publicly accessible Genomics of Drug Sensitivity in Cancer database[17]. Specifically, we collected copy number variation (CNV) profiles of 1,389 cell lines from the Genomics of Drug Sensitivity in Cancer database. CNV, characterized by a copy number change involving a DNA fragment that is ~1 kb or larger, has influence on gene expression levels, phenotypic variation and susceptibility to various disease[18]. We meticulously curated the CNV data for 777 cancer genes from COSMIC[19]. The CNVs were subsequently used as distinctive genomic features of the cell lines, which were then integrated into the DrugBAN model to accomplish the drug response prediction. In the original dataset, the interactions between cell lines and drugs were quantified in terms of IC_{50} values, a continuous variable. In our experimental design, we labelled the drug–cell line pairs by categorizing an IC_{50} value below −3 as sensitive and, conversely, an IC_{50} value exceeding 3 as resistance, aligning with the binary classification framework of the DrugBAN model. To rigorously evaluate the model's robustness, we maintained the consistent network architecture while substituting the original protein data with CNV profiles.

Mutation–drug association prediction using DrugBAN

For mutation–drug association prediction, we undertook a comprehensive collection by screening and integrating an array of mutation–drug associations, which were classified as either sensitive or resistant. These associations were meticulously curated from the distinguished cancer variant interpretation database Meta-KB[20], resulting in a dataset comprising 1,754 clinically evidenced or experimentally verified mutation–drug association pairs, involving 661 missense mutations and 184 drugs. To integrate mutation data into the DrugBAN model, we characterized missense mutations using the tool Seq2Feature[21], deriving 248 distinctive features for each mutation. Consistent with cell line–drug response prediction, the structural integrity of the DrugBAN model remained unaltered. We simply substituted the original protein data with mutation data, assessing the model's adaptability and efficacy in mutation–drug association prediction.

Data availability

The data used in our study are available via GitHub at https://github.com/vshy-dream/DrugBAN-reusability (ref. 22). All of the data used in the original paper by Bai et al. for testing DrugBAN are available via GitHub at https://github.com/peizhenbai/DrugBAN/tree/main/datasets (ref. 23).

Code availability

The original DrugBAN code is available via GitHub at https://github.com/peizhenbai/DrugBAN (ref. 23). Our modified DrugBAN version with changes is available via GitHub at https://github.com/vshy-dream/DrugBAN-reusability (ref. 22).

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Author contributions

Z.Y. conceived and supervised the project. H.S., T.X. and Z.Y. designed the computational experiments and data analyses. H.S. and X.W. prepared the data. H.S., T.X. and W.G. implemented the methods, conducted the experiments and performed data analyses. H.S., T.X. and Z.Y. wrote the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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